Section: **3.4 – METAL ASSEMBLIES**

Chapter: **3340**

Title: **FRAMES**

Revision: **0.4**

Date: **16/05/2019**

1 INTRODUCTION

1.1 OBJECT

This document describes the typical process for performing the static strength check of a fuselage metallic ring frame.

1.2 SCOPE AND LIMITATIONS

Strength analysis of ring frames of fuselages made of metallic material.

Out of the scope of methodologies presented in this chapter:

- ✓ Frames made entirely of composite materials or hybrid composite/metallic frames or fuselage with composite skins are not covered within
- ✓ Bulkhead frames or frames with a deep web where the slender beam or the slightly curved beam conditions (see chapters §3110 "BEAMS SECTIONS" and §3111 "STRESS IN CURVED BEAMS SECTIONS") are not complied
- ✓ Frame sections and vicinities of locations with discrete load introductions or crossings with other beam elements

2 STRUCTURAL DESCRIPTION

Description of typical fuselage and frame components is shown graphically in following figures.

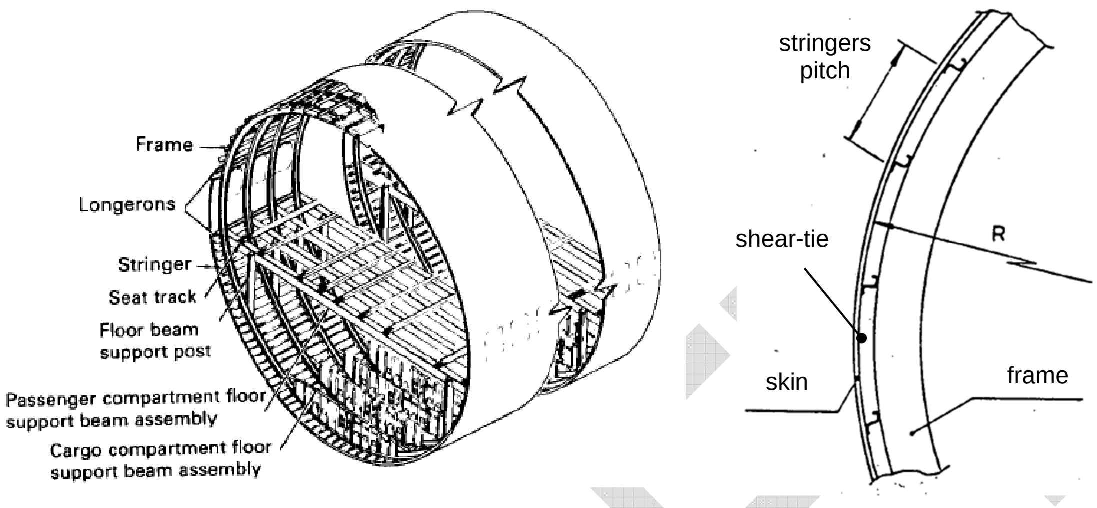

Figure 1. Typical fuselage slice of a commercial transport aircraft and frame lateral view

Standard main elements of a frame assembly:

- Frame
- Shear-tie element, in case of differential frames, between the “floating” frame and the skin
- Skin (forward and rearward subpanels)
- Stabilizers (or clips, or cleats) providing lateral support to frame at certain stations

These elements are attached together by means of fastened joints.

Integral frames include mouse holes for permitting continuous stringers in X-direction. In differential frames, the path for the stringers is available among the “floating” frame and the spaced shear ties.

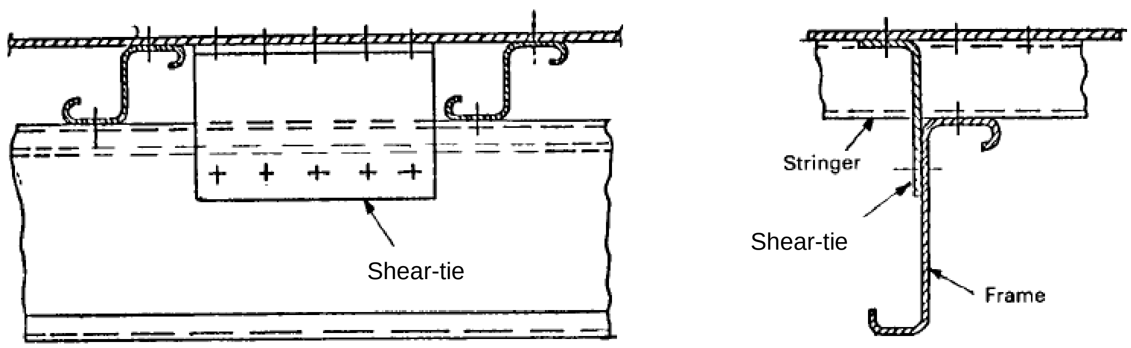

Figure 2. Frame configuration with frame over stringer and shear-ties between stringers (Ref. 1)

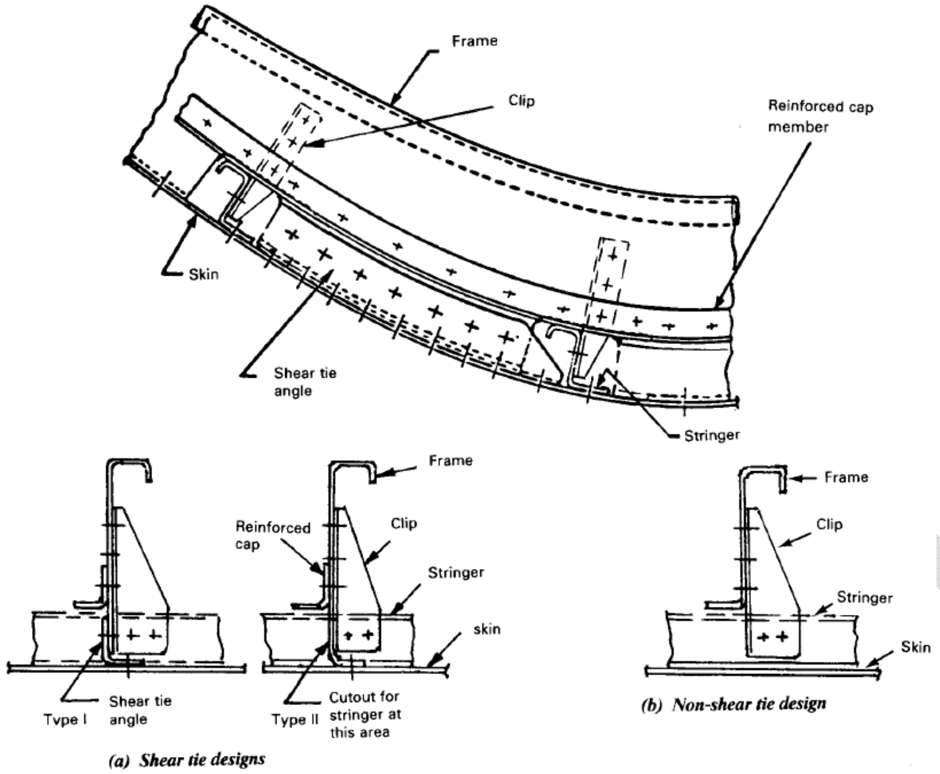

Figure 3. Frame configurations with and without shear-ties (Ref. 1)

3 GENERAL STRESS & STRENGTH ANALYSIS PROCESS

Description of steps to perform the strength check of a metallic frame:

1. Section inertia and pure allowables

   Recap frame geometry, materials ultimate strengths, calculate section inertia properties and rest of pure/uni-axial allowable loads: fastened joints strengths, crippling of section segments (checking flanges effective support), and web buckling shear load

2. Obtain integrated internal loads, i.e. terns (P_x, M_y, V_z) , for all applicable cases (*), typically from a global FEM, at the frame section (as per sub-chapters §4.1 and §4.2).

   (*) It is generally recommended, instead of performing the stress&strength analysis of all resulting cases, to do a reduced selection of the potentially critical ones. There are tools available to do this task with load sets defined by 3 components. It is also possible, for preliminary sizing and optimization phases, to do this selection with the points of the bi-dimensional envelope of pairs (P_x, M_y) plus the case with maximum shear V_z .

For each potentially critical load case…

3. Stress at applied loads, at limit and ultimate load levels, as per sub-chapter §4.3. This is strictly not required for obtaining the static margins, but preferred for better understanding.

4. By scaling each load set by a variable load factor $LF\cdot(P_x, M_y, V_z)$, calculate the stress distributions (as per §4.3) , failure indices (as per sub-chapter §5) and get the *RF* for the section and load case as the *LF* value when the maximum failure index (***) of all possible failure modes is equal, and not greater, than 1.0.

   (**) Failure indices (or damages levels), are preferred here when comparing local/point stresses with strengths for each failure mode. A failure index is defined as the applied stress divided by the allowable stress for a failure mode. This convention is used to differentiate local static margins for a certain load level at section points/details (which are not necessarily proportional to increase of applied forces due to non-linear effects), from the reserve factor, which must be referred to the "gross" applied loads.

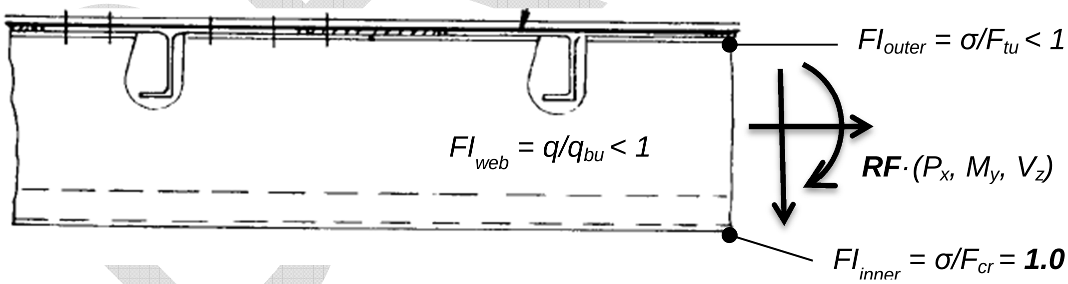

Figure 4. Illustration of differentiation between failure index and reserve factor

4 LOADS AND STRESSES

4.1 INTERNAL LOADS

Stress and strength check model presented in this report requires that internal loads for frame analysis are presented integrated in the section as axial force, vertical shear and bending moment.

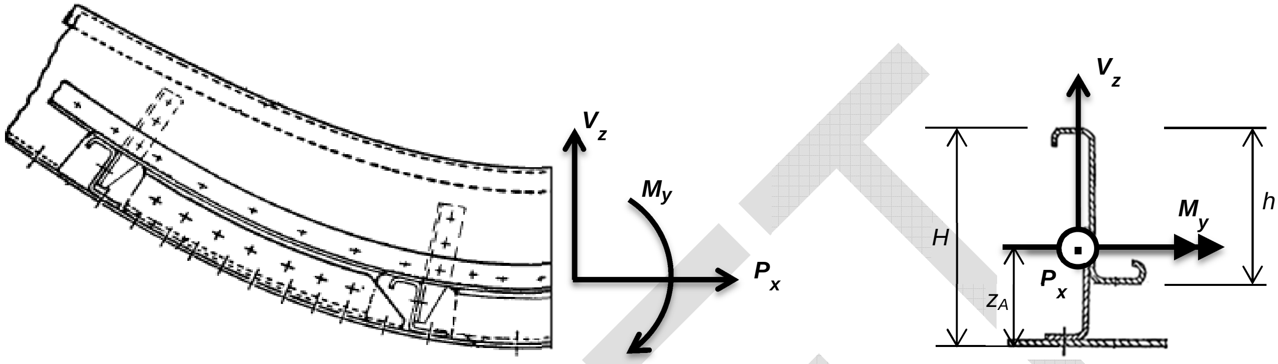

Figure 5. Internal loads integrated at section of analysis

Definitions of internal loads required for stress and strength analysis:

- P_x Axial load force
- V_z Vertical shear force
- M_y Bending moment (positive if causes tension in inner frame cap)
- z_A Height, from outer skin surface (LOFT), of point of loads application

Remarks:

- ✓ Curved beam configuration shall induce radial stresses in the frame web. Theses stresses shall either be directly from GFEM if sufficiently detailed or calculated using curved beams theory from bending loads.
- ✓ The relevant area for integrating internal loads comprises the frame and the halves of the adjacent skin subpanels between the frame of interest and the following ones.

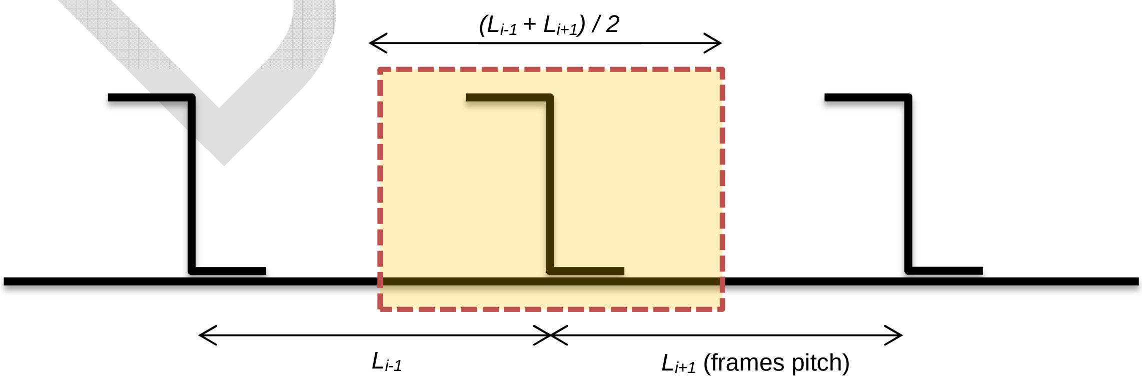

Figure 6. Relevant area for integrating loads for frame section analysis (shaded)

- ✓ Note that skin shear and longitudinal (stringers parallel) loads are not participating in this analysis. Skin role in this analysis is just to permit the analysis of the frame. The full stress and strength analysis of the skin is done within the fuselage stiffened panel method.

4.2 LOADS EXTRACTION FROM GFEM

Internal loads in relevant GFEM elements composing the frame elements shall be integrated at a reference analysis point in the frame section.

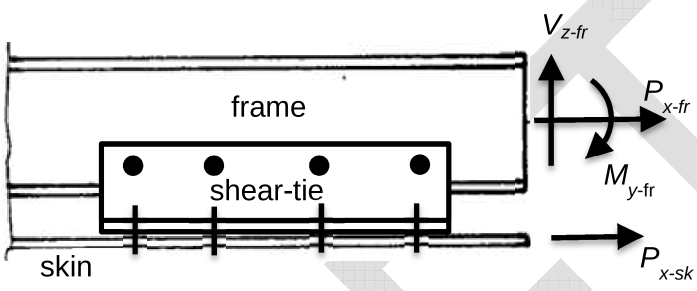

Figure 7. Frame and skin, lateral view

- ✓ Loads in frame

  Axial load ($P_{x\text{-}fr}$), vertical shear ($V_{z\text{-}fr}$), and bending moment at loads extraction point ($M_{y\text{-}fr}$), typically at the centre of gravity of the frame element(s), at a distance (*o*) from the skin/LOFT surface

- ✓ Axial load from forward and rearward skin panels under the frame (fuselage circumferential direction)

  $P_{x\text{-}sk} = (N_x \cdot b_{sk})_{fwd}/2 + (N_x \cdot b_{sk})_{rwd}/2$

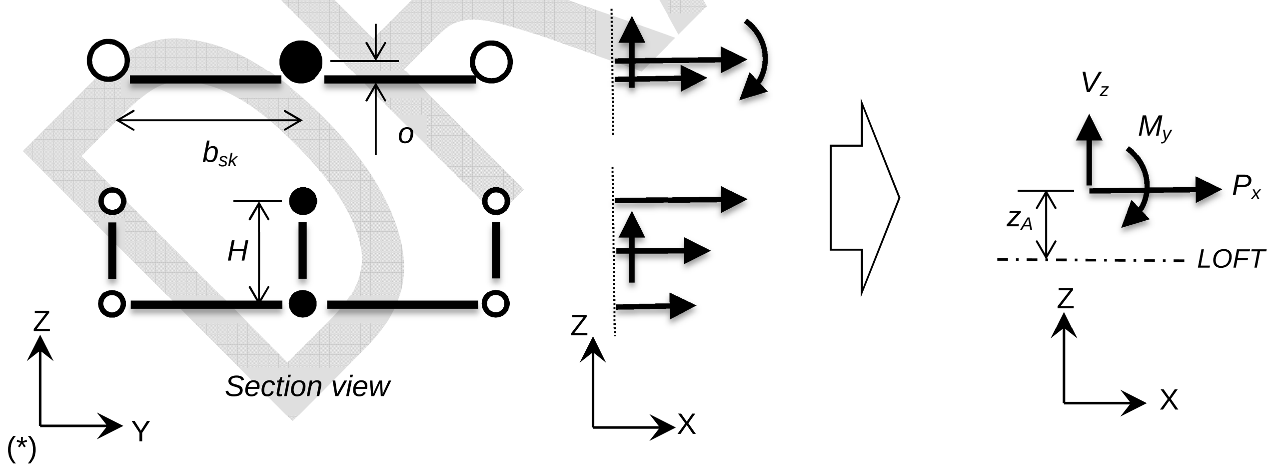

Figure 8. Forces in GFEM frame elements (2 idealizations) to reference point of application

(*) X, Y, Z reference axis system in this sketch is local for this specific frame section analysis. X is aligned with the frame-beam elastic axis at each point, and Z is parallel to radial direction pointing inward. Note the local Y axis will be always parallel to global fuselage longitudinal direction (typ. X axis of the default reference system in the aircraft/fuselage GFEM)

- ✓ Radial loads in web

  When the frame idealization in the GFEM includes 2D elements normal the skin surface, radial stresses/loads shall be directly captured from those elements.

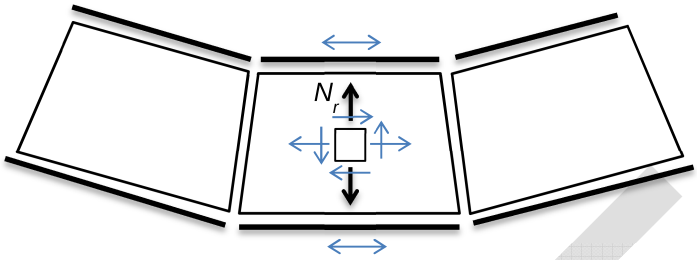

Figure 9. Curved frame in GFEM including 2D elements for the web segment

4.3 STRESS DISTRIBUTION

Stress distribution due to axial and bending loads

For a given set of loads (P_x, M_y) applied at a point z_A , axial stress distribution shall be calculated using classical beams analysis theory as described in chapter §3110 "BEAMS SECTIONS" (taking account plastic bending when relevant) and effective skin width computed as described in chapters §3185 "EFFECTIVE SKIN WIDTH" and §3180 "INTER-RIVET BUCKLING". The crippling stress relevant for the effective width calculation shall be obtained using a piece of the frame or shear-tie riveted to the skin, as shown in figure below (*).

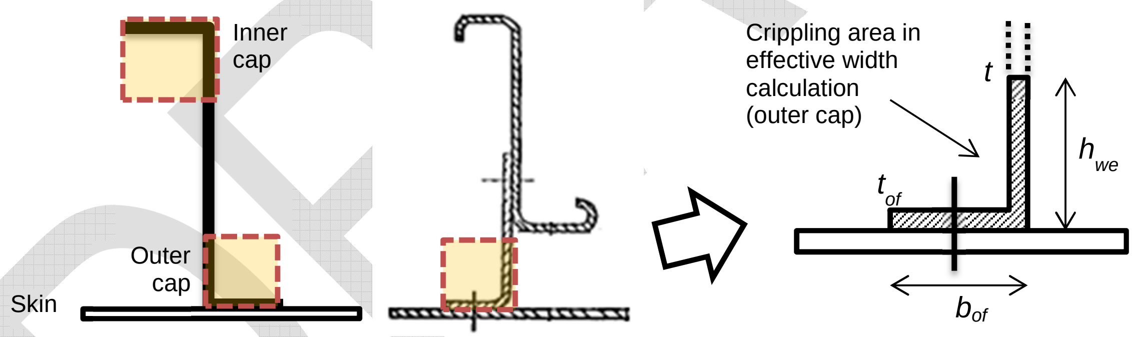

Figure 10. Reference cross section for crippling stress relevant for effective width calculation

(*) For calculating the crippling stress of the outer cap of the frame section, including the flange riveted to the skin (outer flange) and piece of the web, the effective height of the vertical segment to consider shall be:

$$h_{we} = \min\left[\, h/2,\ 0.85 \cdot t \cdot (E/F_{cr\text{-}of})^{0.5} \,\right]$$

…where $F_{cr\text{-}of}$ is the crippling stress of the flange riveted to the skin (outer flange). The crippling stress of the outer cap section is then:

$$F_{cr} = [\, F_{cr\text{-}of} \cdot (t_{of} \cdot b_{of}) + F_{cr\text{-}we} \cdot (t \cdot h_{we}) \,] / (t_{of} \cdot b_{of} + t \cdot h_{we})$$

…where the crippling stress of the vertical flange part is obtained as the maximum between:

- ✓ the value calculated with width equal to the frame height *H*,
- ✓ and the value calculated with the effective width h_{we} and one edge free.

The same approach shall be used later to calculate the crippling stress of the inner cap.

Note effective skin is function of the applied loads (except for cases with skin in tension), so previous problem, in a general case, does not have an explicit closed-form solution, and must be solved numerically (non-linear due material-plasticity and due to effective width dependency of stress in skin). The computational cost of performing these calculations in a fuselage, with hundreds of frame locations and hundreds of loads cases may be significant. A conservative simplification of the section can be done for the analysis at ultimate load level, consisting in reducing the cross section of the frame+skin into two concentrated masses at inner and outer caps (areas defined as explained in previous paragraphs).

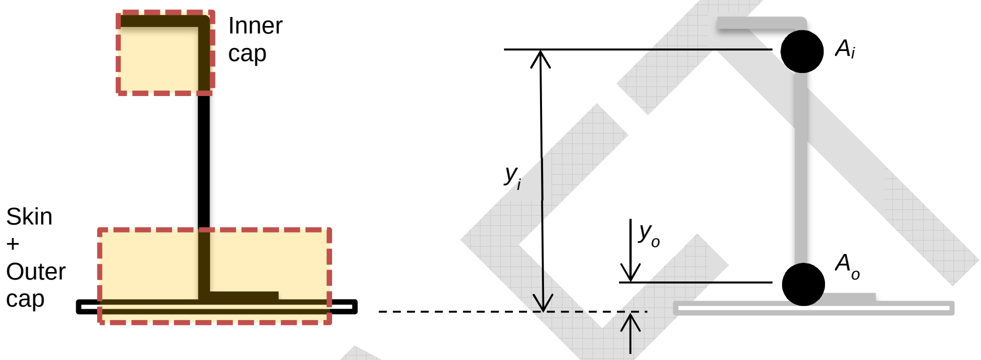

Figure 11. Section idealized as two concentrated masses

The problem is solved simply by stating the equilibrium of axial load and bending moment in the cross section, obtaining the solution as a pair of forces at caps centroids. A further conservative simplification, for removing the numerical iterations remaining in the stress analysis of the outer cap, would be to calculate the effective skin assuming its stress level reached the crippling (or inter-rivet buckling) stress of the cap.

Additional remarks:

- ✓ This method is un-coupled from the full skin analysis. Forced crippling loads induced by the potential development of diagonal tension in the skin is not accounted here. An extra provision (or penalization) of the strength check of the frame/shear-tie cap riveted to the skin may be recommended (TBC).

Shear stress

Shear stress distribution due to vertical shear V_z can be generally approximated by dividing the value of the shear force by the height of the frame web (adding the height of the shear-tie when applicable, see Figure 5), and then by the web thickness:

$q =$ V_z / h … for floating frames and for spans over the frame's stringers mouse-holes

V_z / H … for integral frames or frames with shear ties attached to skin

$\tau = q / t$

For practical reasons this conservative approach is preferred onto a detailed calculation of the shear stress distribution across the section. This mean/uniform value of q shall be used either in the local stability analysis of the frame web and for calculating fastener loads in attachments.

Curved beam correction

The following corrections shall be applied to abovementioned stress distribution to account for curved beam effects:

- Axial stress: No correction applies. Note pure pending and pure axial stress shall be differentiated before this operation.
- Bending stress: Equal to the bending stress calculated for the section as it were part of a straight beam, multiplied by the factor $1/(1-z_{cg}/R)$, with R the curvature radius of the centroidal axis and z_{cg} the coordinate of the point in the local reference of the section with origin in its centre of gravity (Roarks' book chapter §9). Notice the sign convention for this coordinate is that its value will positive when the point is between the centre of gravity and the centre of curvature of the beam, i.e. the point of interest is more inside the fuselage)
- Shear stress: No correction shall be applied. Note only the mean value across the web segment is relevant for checking local buckling stability and riveted joints.
- Radial stress:
  - When the frame idealization in the GFEM includes 2D elements normal the skin surface, radial stresses shall be directly captured from those elements (see Figure 9)
  - Otherwise, an acceptable approximation of the radial stress in the web of flanged beams in bending, where bending is principally supported by differential axial forces in outer and inner flanges, is the following:

$$\sigma_r = P_i / (R \cdot t)$$

… with P_i the axial load at the inner flange, t is the thickness of the web and R the curvature radius at the analysis point. Note P_i and σ_r have consistent sign convention in that formula: positive tension load leads to positive tension radial stress

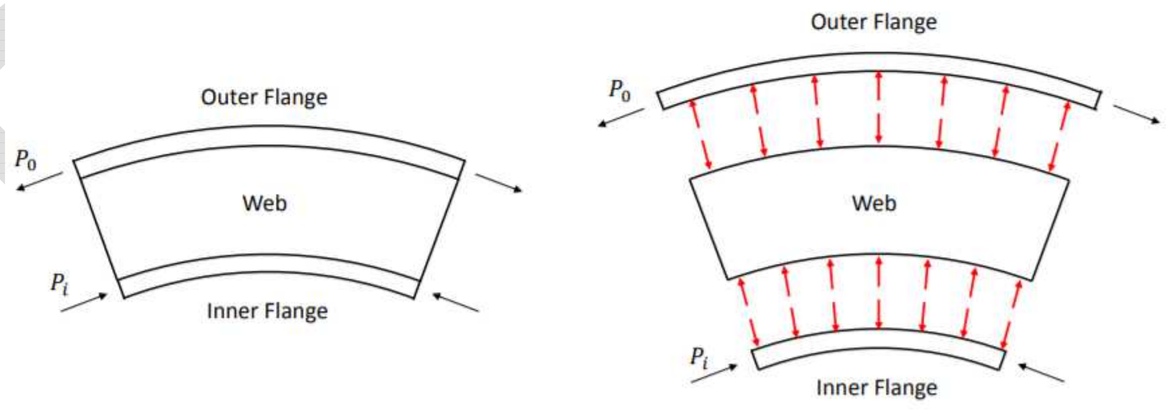

Figure 12. Radial stress in flanged beams

See also discussion in chapter §3111 "STRESS IN CURVED BEAMS SECTIONS".

5 STRENGTH CHECK METHODS

5.1 FRAME

Strength (as stable section)

For each part of the section (*), at most inner and outer points, compare the Von Mises stress with the ultimate tensile material stress F_{tu} at ultimate loads, or with the yield material stress (F_{ty} or F_{cy}) at limit loads (see chapter §3102 "STRENGTH OF METALS").

For locations with fastener lines, the by-pass axial stress to consider in this check shall be calculated using the applicable net-section correction (see *Fastened Joints* sub-chapters).

The static margin for the most critical part-corner shall drive the strength failure margin.

(*) Notice any margin obtained here for the skin is incomplete (shear and stringers-parallel skin loads are not accounted), and shall be completed in complementary stiffened skin analysis.

Crippling strength of caps

If compression and/or bending loads are applied, then some of the flanges or portions of the web may still be under compression and can be crippling-critical.

Method for calculating crippling strengths and flanges-support-effectiveness is described in chapter §3130 "LOCAL BUCKLING AND CRIPPLING OF BEAMS/STRINGERS". If either the frame inner or outer flanges do not provide effective support to the web the section shall be considered failed for any case with compression stresses.

In sections with non-uniform axial stress distribution, as our case, the crippling strength check shall be done segment by segment (not averaging across the whole section as in pure axial cases). The static margin for the most critical segment shall drive the crippling failure margin.

a) If the stress analysis was done idealizing the section in two concentrated masses, the applicable stresses to use are obtained by simply dividing the force in the cap by its area

b) If the stress analysis was done using the beams theory on the complete cross section:

- At the inner flange, compare its crippling strength (one edge free) with the stress computed at that location
- At the frame web, were the stress distribution is not uniform, the recommended approach shall be the following:
  - The allowable crippling stress of the segment shall be obtained normally as if the complete width were in compression (i.e. for a bi-supported segment of width *b*)
  - That allowable stress shall be compared, to obtain the static margin, with a weighted reference stress within the width in compression, calculated as $(0.67 \cdot \sigma_{cmax} + 0.33 \cdot \sigma_{cmin})$. If the neutral axis crosses the segment, that is $0.67 \cdot \sigma_{cmax}$.

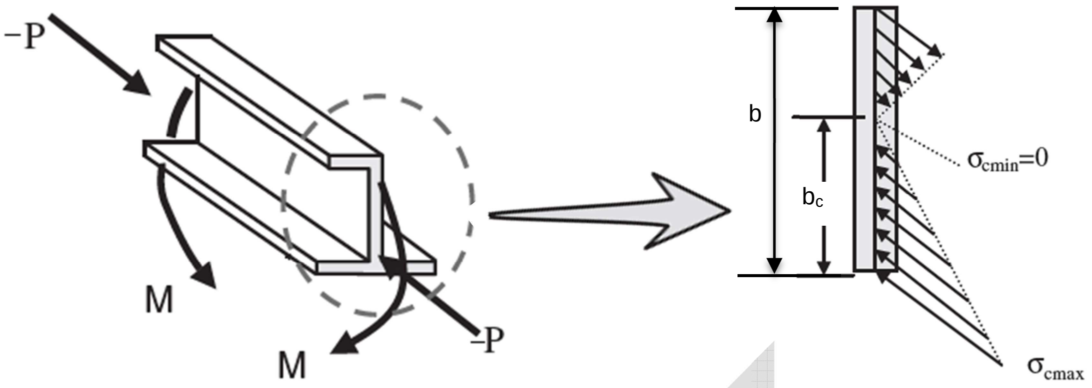

Figure 13. Width effective for crippling analysis in segments with non-uniform stress

Web Stability

Generally, shear driven buckling is not allowed in frame sections (however this depends on the program policy).

Local buckling of web under combined compression, shear, and bending, supported on inner and outer flanges, shall be checked using method described in chapter §3150 "PLATE/SHELL BUCKLING" . In particular, formulas in Niu's book (Ref. 2) in chapters §11.3(A)(B)(C) to obtain the uni-axial buckling elastic allowable, §11.2(B) to get the plasticity correction and §11.5 to get the static margin in combined loads.

The web plate shall be assumed straight (rectangular) with a high aspect ratio (a/b >> 1), or using the real stringers pitch as length if vertical stabilizers (clips) at crossings are riveted to the frame web.

Apply corrections in §3165 "HOLED WEBS" if lightening holes are present.

Lateral Buckling

Check against lateral buckling of the frame shall be done according methodology described in chapter §3170 "LATERAL STABILITY OF STRINGERS/FRAMES".

If the inner flange of a frame is riveted to an inner structure this failure mode is not possible and this check shall not be done.

5.2 SHEAR-TIE

Strength

To be performed as explained for the frame. In this case, typically, as the shear-tie is not considered effective to transfer axial and bending forces at frame section, it will be only sustaining shear stress.

Stability

A buckling analysis of the shear-tie vertical segment in shear, bi-supported in flange riveted to skin and at frame riveting line, shall be performed, assuming a very high aspect ratio (a/b >> 1).

5.3 SKIN

As previously mentioned, the skin contribution to this analysis is mainly to account its contribution in the stress distribution of the frame, by means of the effective-skin model. It is not intended in this method to present realistic static margins of the skin.

The strength and stability analysis of the skin shall be fully described in the stiffened skin method described in chapter §3320 “CURVED STIFFENED PANEL” .

5.4 FASTENED JOINTS

Frame to Shear-Tie and Shear-Tie (or Frame) to Skin

Applicable loads:

- ✓ The mean shear flow in the web shall be used in this analysis:

  $$q = V_z / H$$

  Shear forces transferred by the rivets are then obtaining multiplying this load by the rivets pitch and dividing by the number of existing rivets rows at each of the attachments. Preferably if pitch is not really uniform, multiplying the mean shear flow by the stringers spacing and dividing by the number of rivets along that frame span.

  $$Q_i = q \cdot p / n_{rows} \quad \text{or} \quad Q_i = q \cdot w / n_{rivets}$$

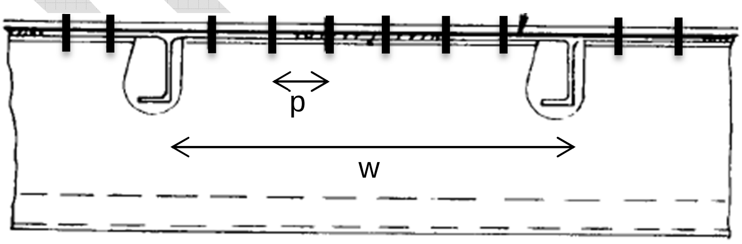

Figure 14. Longitudinal fastened joint between frame and skin

- ✓ The axial by-pass stress shall be according the calculated stress distribution in axial and bending loads

The fastened joint strength shall be checked according §3220 “FASTENER FAILURE” and §3230 “PLATE BEARING AND PULL-THROUGH”.

5.5 STABILIZERS / CLIPS AT STRINGERS CROSSINGS

To Be completed.

Reserved to explain conditions that shall be verified to assure effective support for lateral buckling of frames.

6 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	*Units*	*Definition*
b	mm	Width (typically width of segment in frame section)
FI	-	Failure index, i.e. stress at a point/detail divided by the corresponding allowable stress for a failure mode
F_{cy}	MPa	Yield compression stress
F_{cr}	MPa	Crippling stress
F_{ty}	MPa	Yield tensile stress
F_{tu}	MPa	Ultimate tensile stress
H	mm	Total height of the frame section, from outer skin surface to most inner point for the frame
h	mm	Height of the frame part in the section (not including shear-tie elements of height of mouse holes)
L	mm	Frames pitch, i.e. distance between consecutive frames
M_y	Nmm	Bending moment
p	mm	Fasteners pitch
P_x	N	Axial force
q	N/mm	Shear load flow in frame web
Q	N	Shear force transferred by a rivet
V_z	N	Shear load (force) in frame section
w	mm	Stringers pitch, typically equal to the length of the unit frame spam relevant for analyses
σ	MPa	Axial stress at a point of the section
τ	MPa	Shear stress in frame web
RF	-	Reserve Factor, i.e. allowable load divided by applied load, or times the applied load has to be applied to get the first failure at the frame
z_A	mm	Distance from loads application point to outer skin surface

REFERENCES

	Document Reference	*Issue*	*Title*
Ref. 1	[M.C.Y. Niu]	1st Ed.	AIRCRAFT STRUCTURAL DESIGN
Ref. 2	[M.C.Y. Niu]	2nd Ed.	AIRFRAME STRESS ANALYSIS AND SIZING
Ref. 3	[W. C. Young & R. G. Budynas]	7th Ed.	ROARK'S FORMULAS FOR STRESS AND STRAIN
Ref. 4	[C. Kassapoglou]	2nd Ed.	Design and Analysis of Composite Structures

SOFTWARE

Program	***Version***	***Description***

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

Revision Date	***Authors***	***Change Reason***	***Pages/Sub-chapters affected***
0.4 16/05/2020	G. Baños	First issue	All pages

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	***Getafe***	***Manching***
Stress Methods (TEASP-TL3)	G. Baños 16/05/2020	Florian Glock XX/XX/2020

Table 3. *Approval signatures table for last revision of this chapter*